Weili Xu

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Department of Electrical & Computer Engineering, University of Illinois Urbana-Champaign

Research Profile

Senior undergraduate in Computer Engineering in the UIUC–Zhejiang University dual-degree program. **Seeking Ph.D. opportunities in machine learning systems (MLSys) starting in Fall 2027.** Research interests: agentic LLM serving, hardware-aware sequence modeling, and efficient long-context multimodal systems.

Education

University of Illinois Urbana-Champaign Aug. 2023 – Present
B.S. in Computer Engineering; GPA: 3.97/4.00 *Urbana, IL*

Zhejiang University Aug. 2023 – Present
B.S. in Computer Engineering (Dual Degree); GPA: 4.20/4.30 *Hangzhou, China*

Selected coursework (all A/A+): ECE 408 Applied Parallel Programming; ECE 391 Computer Systems Engineering.

Research and Industry Experience

Together AI May 2026 – Sep. 2026
Research Intern, Frontier Performance Team *San Francisco, CA*

- Optimized LLM inference infrastructure for long-context and high-throughput agentic workloads.

University of Illinois Urbana-Champaign Aug. 2025 – Present
Research Intern in Machine Learning Systems *Urbana, IL*

- **ThunderAgent (co-first author; ICML 2026 Spotlight):** developed program-aware scheduling for agentic inference, addressing KV-cache thrashing and cross-node memory imbalance.
- Achieved 1.5–3.6× serving throughput and 1.8–3.9× RL rollout throughput on agentic workloads on an 8×H100 node.

Zhejiang University Aug. 2023 – Jul. 2025
Research Intern in Machine Learning *Hangzhou, China*

- **AuroraLong (first author; ICCV 2025):** developed linear-RNN-based multimodal modeling for hour-long video understanding; achieved 8.13× faster inference on an RTX 3090 GPU.
- **Video-MMLU (ICCV Workshops 2025):** developed a multi-discipline lecture-understanding benchmark evaluating 90+ video LLMs; Outstanding Paper Award and Oral Presentation at the Knowledge-Intensive Multimodal Reasoning Workshop.

Open-Source Contributions

TileGym (NVIDIA's cuTile Kernel Library) Dec. 2025 – Jun. 2026
Open-Source Code Contributor *Remote*

- Implemented backward-pass FlashAttention and SwiGLU kernels in cuTile with numerical correctness tests and benchmarking; achieved >10% higher throughput than the optimized Triton baseline.

Teaching

Zhejiang University–University of Illinois Urbana-Champaign Institute
Undergraduate Teaching Assistant Aug. 2024 – May 2025
ECE 220 Computer Systems & Programming (Spring 2025); ECE 120 Introduction to Computing (Fall 2024).

Publications

* Equal contribution. † Project lead.

1. **Evolving Infra: Agent-Adaptive LLM Serving at Scale**
   Hao Kang*, **Weili Xu***, Ziyang Li*, Zheng Du*, Hanchen Li, Tian Xia, Song Bian, Tushar Krishna, Huizi Mao, Ion Stoica, Song Han.
   Technical report, 2026. arXiv:2611.12304.

2. **FlashStates: Unlocking Flexible Hybrid State Management for Agentic LLM Inference**
   Chenxi Li*, **Weili Xu***,†, Tianyi Zhang, Jue Wang, Zhongzhu Zhou, Chenfeng Xu.
   Preprint, 2026.

3. **ThunderAgent: A Simple, Fast and Program-Aware Agentic Inference System**
   Hao Kang*, Ziyang Li*, **Weili Xu***, Xinyu Yang*, Yinfang Chen, Junxiong Wang, Beidi Chen, Tushar Krishna, Chenfeng Xu, Simran Arora.
   ICML 2026, Spotlight.

4. **Bringing RNNs Back to Efficient Open-Ended Video Understanding**
   Weili Xu, Enxin Song, Wenhao Chai, Xuexiang Wen, Tian Ye, Gaoang Wang.
   ICCV 2025, pp. 23453–23465.

5. **Video-MMLU: A Massive Multi-Discipline Lecture Understanding Benchmark**
   Enxin Song, Wenhao Chai, **Weili Xu**, Jianwen Xie, Yuxuan Liu, Gaoang Wang.
   ICCV Workshops 2025, pp. 6099–6113.

Recent Research

Evolving Infra — Co-first author. Agent-adaptive LLM serving that decouples compute allocation from KV-cache management, enabling online prefill/decode adaptation while retaining reusable agent context.

FlashStates — Co-first author and project lead. Flexible hybrid state management combining arbitrary recurrent-state checkpointing, dependency-aware KV/state management, and cost-aware eviction for agentic inference.

Technical Skills

Programming: Python, C/C++, Triton, CUDA, cuTile, Java, RISC-V Assembly.

Frameworks and systems: PyTorch, vLLM, SGLang, TensorRT-LLM, Dynamo, NIXL, Mooncake, slime, Megatron-LM.

Tools: Git, Linux/UNIX, Slurm, Kubernetes, Docker, GDB, PDB, Nsight Systems, Nsight Compute.